ASSIGNMENT 02 · CO3 + CO4

Reconstruct or Create?

A comparative study of four Generative AI models

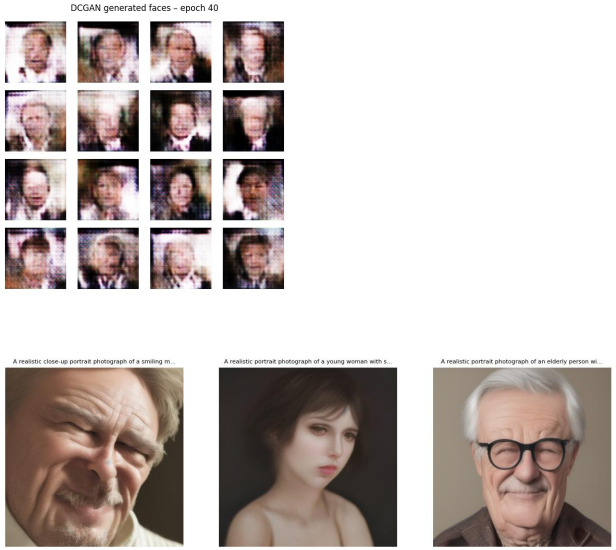
Autoencoder · VAE · DCGAN · Diffusion

Generative AI · MIT Academy of Engineering, Pune

SUBMITTED BY

Sanskar Ramesh Meharkar 202401110029
Meghana Shirish Innani 202401110009
Dewesh Barapatre 202401110053

Dataset: LFW faces, 64×64 · Code: Google Colab notebook



DCGAN samples (left) and pretrained diffusion portraits (below)

All images shown were generated by our DCGAN or the diffusion model.

Four models, two jobs

RECONSTRUCTION · CO3

GENERATION · CO4

Autoencoder

Squeezes the image into 256 numbers, then rebuilds it (MSE loss)

+ Simple, fast, stable

— Blurry, loses detail

Used for: Compression, denoising

VAE

Learns a distribution (μ, σ), samples z = μ + σ·ε (recon + KL loss)

+ Can sample new images

— Even softer outputs

Used for: Representation learning

DCGAN

Generator fakes images from noise; discriminator judges real vs fake

+ New images from our data

— Unstable, artefacts

Used for: Image synthesis, augmentation

Diffusion

Removes noise step by step, guided by a text prompt (U-Net)

+ Best quality + prompt control

— Heavy compute

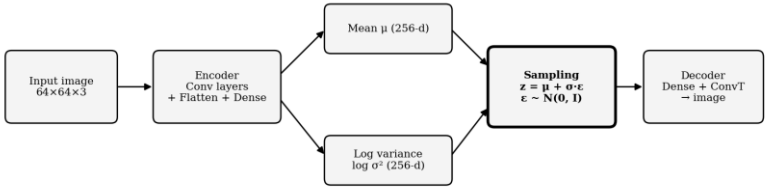
Used for: Text-to-image, design

Inside each model

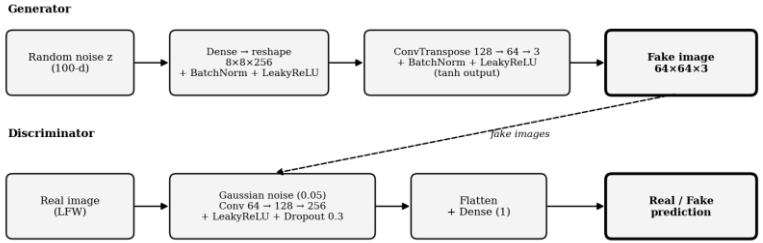
Autoencoder: compress → rebuild



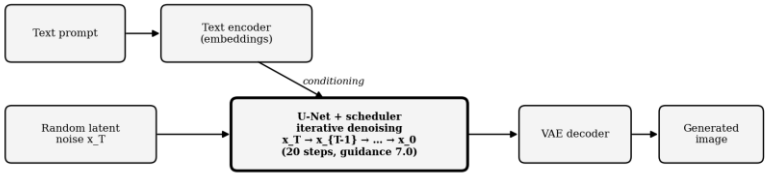
VAE: learn a distribution, then sample



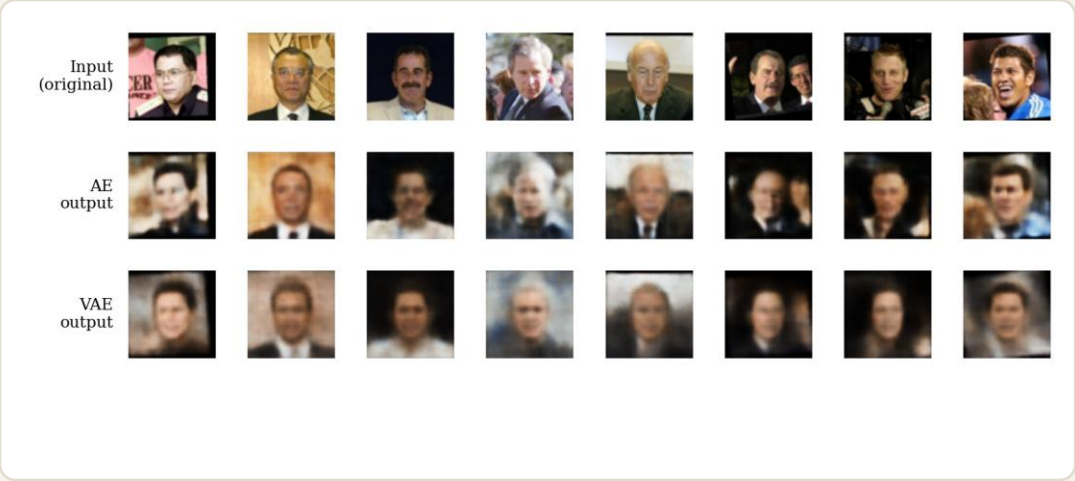
DCGAN: generator vs discriminator



Diffusion: denoise step by step, guided by text



Activity A: AE keeps more detail than VAE



Notebook output · same 8 input images from the 400 validation faces

VAE sampled faces are blurry and low-detail but face-like: the latent space can be sampled.

0.0103
AE validation MSE

−79%
val. MSE drop in 15 epochs

Validation set (400 images)

AE MSE 0.0103 · PSNR 20.24 dB · SSIM 0.599
VAE MSE 0.0234 · PSNR 16.60 dB · SSIM 0.419

- AE keeps colour, layout and rough shape
- Fine texture and edges get blurred
- VAE faces look smoother and “averaged”
- Main difference: loss of detail, not distortion

Activity B: diffusion looks far more realistic

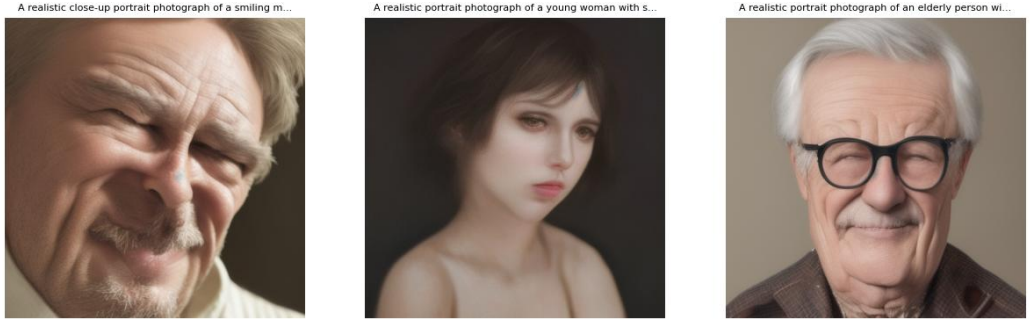


DCGAN · TRAINED BY US

40 epochs · 64×64 · final G 0.89 / D 1.28

Prompt: "A realistic portrait photograph of an elderly person with glasses, neutral background, high detail"

VS



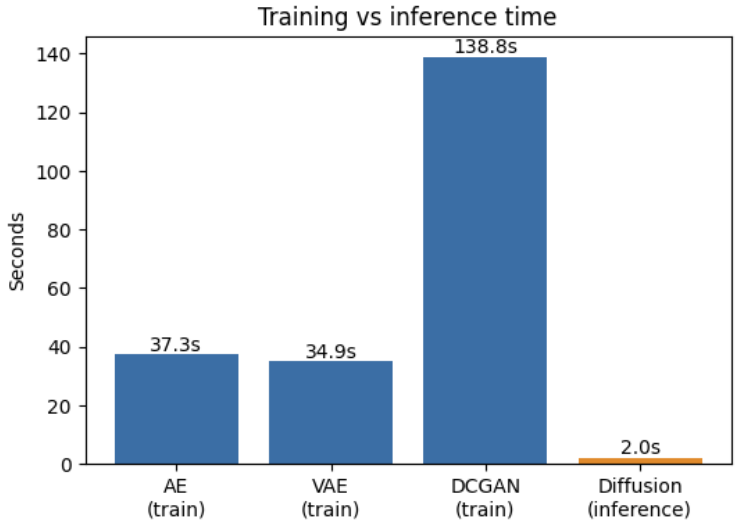
DIFFUSION · PRETRAINED

segmind/tiny-sd · 20 steps · guidance 7.0 · 2.03 s per image

What we saw

DCGAN: new, varied faces from noise, but blurry with grid-like artefacts. Losses stayed near chance (D ≈ 1.3–1.4), yet the images kept improving: judge by eye, not by loss. Diffusion: sharp 512×512 portraits with small artefacts.

DCGAN costs the most; losses are not comparable



3.7–4.0×

DCGAN training time vs AE / VAE

2 nets

trained every step

Model	Loss and scale	Final
AE	MSE, averaged per pixel	0.0103
VAE	BCE + KL, summed	6,662.9 (val.)
DCGAN	Adversarial BCE	G 0.89 / D 1.28
Diffusion	Pretrained	–

Different loss functions, different scales: compare the images, not the numbers.

What we learned and how we used AI

AI usage and verification

- Claude for concepts, notebook debugging and the report
- Every number checked against the executed notebook output
- Theory cross-checked with original papers and docs

Key learning

- AE vs VAE: reconstruction detail vs organised latent space
- GAN: loss values alone do not indicate image quality
- Diffusion: high-quality, prompt-guided generation
- Training time and inference time are different

Contributions

- Sanskar: AE and VAE experiments
- Meghana: DCGAN and diffusion experiments
- Dewesh: comparison table, AI documentation, report and PPT

Conclusion AE/VAE for reconstruction · DCGAN for dataset-specific generation · Diffusion for the best quality at the highest cost